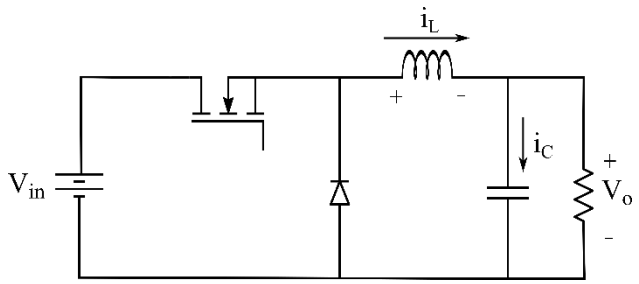
AENL 226 Power Electronics and power systems (Aug-Dec 2026)

Problem sheet 2



Problem 1:

An ideal buck converter must reduce a 48 V input to 12 V. Determine the required duty ratio. If the switching period is 20 μs, also calculate the switch ON time.

Solution:

$D = V_o/V_{in} = 12/48 = 0.25$.

The ON time is $t_{on} = DT_s = 0.25 \times 20\ \mu s = 5\ \mu s$.

Problem 2:

An ideal buck converter has an input voltage of 30 V, a duty ratio of 0.60, and a resistive load of 9 Ω. Find the output voltage, load current, and output power. What are the voltage and current ratings of the power devices?

Solution:

$V_o = DV_{in} = 0.60 \times 30 = 18$ V.

$I_o = V_o/R = 18/9 = 2$ A.

$P_o = V_oI_o = 18 \times 2 = 36$ W.

Problem 3:

An ideal buck converter operates in CCM with $V_{in} = 36$ V, $V_o = 12$ V, switching frequency $f_s = 100$ kHz, and average load current $I_o = 4$ A. Design the minimum inductance so that the peak-to-peak inductor-current ripple does not exceed 20% of the average load current. Also find the peak and minimum inductor currents.

Solution:

$D = V_o/V_{in} = 12/36 = 1/3$.

The maximum allowed ripple is $\Delta I_l = 0.20 \times 4 = 0.8$ A.

For a buck converter, $\Delta I_l = (V_{in} - V_o)D/(Lf_s)$. Therefore,

$L = (36 - 12)(1/3)/(0.8 \times 100{,}000) = 100\ \mu H$.

$I_l,peak = I_o + \Delta I_l/2 = 4 + 0.4 = 4.4$ A.

$I_l,min = I_o - \Delta I_l/2 = 4 - 0.4 = 3.6$ A.

Problem 4:

A buck converter has V_{in} = 24 V, V_o = 8 V, L = 80 μH, and f_s = 50 kHz. It operates in CCM with negligible capacitor ESR. Determine the peak-to-peak inductor-current ripple and the minimum output capacitance required to limit the peak-to-peak output-voltage ripple to 40 mV.

Solution:

$D = V_o/V_{in} = 8/24 = 1/3$.

$\Delta I_l = (V_{in} - V_o)D/(Lf_s)$

$= (24 - 8)(1/3)/(80 \times 10^{-6} \times 50{,}000) = 1.333$ A.

With negligible ESR, $\Delta V_o \approx \Delta I_l/(8f_sC)$. Hence,

$C \geq \Delta I_l/(8f_s\Delta V_o)$

$= 1.333/(8 \times 50{,}000 \times 0.040) = 83.3$ μF.

Problem 5:

An ideal buck converter has V_{in} = 50 V, V_o = 20 V, L = 200 μH, f_s = 25 kHz, and load resistance R = 10 Ω. Determine: (a) the duty ratio, (b) the inductor-current ripple, (c) the critical load resistance value when I_l,min =0, and (d) the maximum switch current and maximum switch blocking voltage.

Solution:

(a) $D = V_o/V_{in} = 20/50 = 0.40$.

(b) $\Delta I_l = (V_{in} - V_o)D/(Lf_s)$

$= (50 - 20)(0.40)/(200 \times 10^{-6} \times 25{,}000) = 2.4$ A.

(c) $I_o = V_o/R = 20/10 = 2$ A. Thus,

I_l,min $= I_o - \Delta I_l/2 = 2 - 1.2 = 0.8$ A.

(c) If load resistance changes, then average inductor current also changes. If I_l,min = 0, then I_o,crit $= \Delta I_l/2 = 1.2$ A. Therefore, Rcrit $= V_o/I_o$,crit $= 20/1.2 = 16.67$ Ω.

(d) I_l,peak $= I_o + \Delta I_l/2 = 2 + 1.2 = 3.2$ A. For an ideal buck converter, the switch must block approximately V_{in} when OFF, so Vswitch,max = 50 V.

Problem 6:

An ideal boost converter must raise 18 V to 30 V. Determine the duty ratio. If the switching frequency is 50 kHz, find the switch ON time.

Solution:

$D = 1 - V_{in}/V_o = 1 - 18/30 = 0.40$.

$T_s = 1/f_s = 1/50{,}000 = 20$ μs.

$t_{on} = DT_s = 0.40 \times 20 = 8$ μs.

Problem 7:

An ideal boost converter raises 10 V to 25 V and supplies a 50 Ω load. Find the duty ratio, output current, output power, and average input current.

Solution:

$D = 1 - 10/25 = 0.60$.

$I_o = V_o/R = 25/50 = 0.50$ A.

$P_o = V_o I_o = 25 \times 0.50 = 12.5$ W.

For an ideal converter, $P_{in} = P_o$, so $I_{in} = P_o/V_{in} = 12.5/10 = 1.25$ A.

Problem 8:

An ideal boost converter operates in CCM with $V_{in} = 24$ V, $V_o = 48$ V, switching frequency $f_s = 100$ kHz, and output power $P_o = 120$ W. Select the minimum inductance so that the peak-to-peak inductor-current ripple is no more than 20% of the average input current. Also find the peak and minimum inductor currents.

Solution:

$D = 1 - 24/48 = 0.50$.

$I_l,avg = I_{in} = P_o/V_{in} = 120/24 = 5$ A.

Allowed $\Delta I_l = 0.20 \times 5 = 1$ A.

For a boost converter, $\Delta I_l = V_{in}D/(Lf_s)$. Thus,

$L = 24 \times 0.50/(1 \times 100{,}000) = 120$ μH.

$I_l,peak = 5 + 1/2 = 5.5$ A; $I_l,min = 5 - 1/2 = 4.5$ A.

Problem 9:

An ideal boost converter has $V_{in} = 15$ V, $V_o = 30$ V, output power $P_o = 60$ W, and switching frequency $f_s = 50$ kHz. Neglect capacitor ESR. Find the minimum output capacitance required to limit the peak-to-peak output-voltage ripple to 1% of V_o.

Solution:

$D = 1 - 15/30 = 0.50$.

$I_o = P_o/V_o = 60/30 = 2$ A.

Allowed $\Delta V_o = 0.01 \times 30 = 0.30$ V.

For a CCM boost converter, $\Delta V_o \approx I_o D/(f_s C)$. Therefore,

$C \geq I_o D/(f_s \Delta V_o) = 2 \times 0.50/(50{,}000 \times 0.30) = 66.7$ μF.

Problem 10:

A practical boost converter raises 20 V to 50 V and supplies a 100 Ω load. Its efficiency is 90%. Determine the output current, output power, input power, average input current,

and the approximate duty ratio if the voltage-conversion relation is assumed ideal.

Solution:
$I_o = V_o/R = 50/100 = 0.50$ A.
$P_o = V_o I_o = 50 \times 0.50 = 25$ W.
$\eta = P_o/P_{in}$, so $P_{in} = 25/0.90 = 27.78$ W.
$I_{in} = P_{in}/V_{in} = 27.78/20 = 1.389$ A.
$D \approx 1 - V_{in}/V_o = 1 - 20/50 = 0.60$.

Problem 11:

An ideal inverting buck–boost converter operates in continuous conduction mode (CCM) with $V_{in} = 24$ V and $V_o = -36$ V. It delivers 72 W to the load at a switching frequency of 100 kHz. Design the minimum inductance so that the peak-to-peak inductor-current ripple is limited to 25% of the average inductor current. Also determine the duty ratio, load resistance, average inductor current, peak and minimum inductor currents, maximum switch current, and maximum voltage stress across the switch and diode.

Solution:
For an ideal inverting buck–boost converter, $|V_o|/V_{in} = D/(1 - D)$. Therefore,
$D = |V_o|/(V_{in} + |V_o|) = 36/(24 + 36) = 0.60$.
The output current is $I_o = P_o/|V_o| = 72/36 = 2$ A.
The load resistance is $R = |V_o|/I_o = 36/2 = 18\ \Omega$.
In CCM, $I_o = (1 - D)I_{l,avg}$. Hence,
$I_{l,avg} = I_o/(1 - D) = 2/0.40 = 5$ A.
The permitted ripple is $\Delta I_l = 0.25 \times 5 = 1.25$ A.
During the switch ON interval, $\Delta I_l = V_{in}D/(Lf_s)$. Thus,
$L = V_{in}D/(\Delta I_l f_s) = 24 \times 0.60/(1.25 \times 100{,}000) = 115.2\ \mu$H.
$I_{l,peak} = I_{l,avg} + \Delta I_l/2 = 5 + 0.625 = 5.625$ A.
$I_{l,min} = I_{l,avg} - \Delta I_l/2 = 5 - 0.625 = 4.375$ A.
The maximum switch current equals the peak inductor current: $I_{switch,max} = 5.625$ A.
For the ideal inverting buck–boost converter, the OFF-state switch and reverse-biased diode each withstand $V_{in} + |V_o| = 24 + 36 = 60$ V.

Problem 12:

An ideal inverting buck–boost converter has $V_{in} = 18$ V, duty ratio $D = 0.55$, inductance $L = 150\ \mu$H, switching frequency $f_s = 50$ kHz, and load resistance $R = 30\ \Omega$. Determine: (a) the output voltage, (b) output current, (c) average inductor current, (d) peak-to-peak inductor-current ripple, (e) whether the converter operates in CCM or DCM, (f) the critical load resistance at the CCM/DCM boundary, (g) the minimum output

capacitance required to limit the peak-to-peak output-voltage ripple to 1% of $|V_o|$, and (h) the peak switch current.

Solution:

(a) $V_o = -V_{in}D/(1 - D) = -18 \times 0.55/0.45 = -22$ V.

(b) $I_o = |V_o|/R = 22/30 = 0.733$ A.

(c) $I_l,avg = I_o/(1 - D) = 0.733/0.45 = 1.630$ A.

(d) $\Delta I_l = V_{in}D/(Lf_s) = 18 \times 0.55/(150 \times 10^{-6} \times 50{,}000) = 1.32$ A.

(e) $I_l,min = I_l,avg - \Delta I_l/2 = 1.630 - 0.660 = 0.970$ A. Since $I_l,min > 0$, the converter operates in CCM.

(f) At the boundary, $I_l,avg,crit = \Delta I_l/2$. Using $Rcrit = 2Lf_s/(1 - D)^2$,

$Rcrit = 2 \times 150 \times 10^{-6} \times 50{,}000/(0.45)^2 = 74.1\ \Omega$.

Because $R = 30\ \Omega$ is less than Rcrit, the CCM result is confirmed.

(g) The allowed ripple is $\Delta V_o = 0.01 \times 22 = 0.22$ V. Neglecting capacitor ESR, $\Delta V_o \approx I_oD/(f_sC)$. Therefore,

$C \geq I_oD/(f_s\Delta V_o) = 0.733 \times 0.55/(50{,}000 \times 0.22) = 36.7\ \mu F$.

A practical standard choice is 47 μF or larger.

(h) $I_l,peak = I_l,avg + \Delta I_l/2 = 1.630 + 0.660 = 2.29$ A. Therefore, the peak switch current is 2.29 A.